Arithmetic Sequences

Algebra 1 · Chapter 4

Handout · 5 topics · 2 easy, 2 medium, 1 hard each
25 practice problems · fully detailed answer key

Name: _____ **Date:** _____

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The Handout *(read first)*

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Detailed Answer Key *(at the end)*

Show all of your work in the space provided. For every sequence problem, write down $A(1)$ and d first — those two numbers determine everything else.

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The Handout

Everything in 4-7, in the order it was taught

What makes a sequence *arithmetic*

A sequence is **arithmetic** when you get from each term to the next by **adding the same number every time**. That number is the **common difference**, written d.

$$70 \overset{+7}{\rightarrow} 77 \overset{+7}{\rightarrow} 84 \overset{+7}{\rightarrow} 91, \cdots$$

To test a sequence, subtract consecutive terms:

$$d = A(2) - A(1) = A(3) - A(2) = A(4) - A(3) = \cdots$$

If all of those differences are equal, it *is* arithmetic. If even one is different, it is **not**.

☞ Once we determine it *is* an arithmetic sequence . . .

Sequence: 70, 77, 84, 91, . . .

Step 1 — find the first term.

$$A(1) = 70$$

Step 2 — write how the second term comes from the first.

$$A(2) = A(1) + 7$$

Step 3 — generalise. This is the *recursive formula.*

$$\begin{array}{ccc} A(2) & = & A(1) + 7 \\ \downarrow & & \downarrow \\ A(n) & = & A(n-1) + 7 \end{array}$$

$$\boxed{A(1) = 70, \qquad A(n) = A(n-1) + 7, \quad n \geq 2}$$

Why $n \geq 2$? The formula tells you how to get a term from the *one before it*. There is no term before $A(1)$, so the rule only makes sense from the second term onward.

☞ Your turn — same three steps

Sequence (a): 3, 9, 15, 21, . . .

$$3 \overset{+6}{\rightarrow} 9 \overset{+6}{\rightarrow} 15 \overset{+6}{\rightarrow} 21, \cdots$$

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Step 1.	First term	$A(1) = 3$
Step 2.	Second term	$A(2) = A(1) + 6$
Step 3.	Recursive formula	$A(n) = A(n-1) + 6, \quad n \geq 2$

★ Where the explicit formula comes from

The recursive formula is fine for finding the *next* term — but to find $A(100)$ you would have to compute all 99 terms before it. Here is how to get a formula that jumps straight there.

Write out the chain of recursive statements and add them all up.

$$\begin{aligned} A(1) &= A(1) \\ A(2) &= A(1) + 7 \\ A(3) &= A(2) + 7 \\ A(4) &= A(3) + 7 \\ \cancel{A(5) = A(4) + 7} & \end{aligned}$$

$$A(1) + A(2) + A(3) + A(4) + A(5) \;=\; A(1) + \Big[A(1) + A(2) + A(3) + A(4)\Big] + \underbrace{7 + 7 + 7 + 7}_{4 \text{ sevens}}$$

Now subtract $A(1)$, $A(2)$, $A(3)$ and $A(4)$ from both sides. Almost everything cancels:

$$\cancel{A(1)} + \cancel{A(2)} + \cancel{A(3)} + \cancel{A(4)} + A(5) = A(1) + \cancel{A(1)} + \cancel{A(2)} + \cancel{A(3)} + \cancel{A(4)} + 4 \cdot 7$$

$$\boxed{A(5) = A(1) + 4 \cdot 7}$$

Read the pattern. To reach the **5th** term you added 7 exactly **4** times. To reach the nth term you add d exactly $n - 1$ times:

$$\begin{array}{ccccc} A(5) & = & A(1) & + \; 4 & \cdot 7 \\ \downarrow & & & \downarrow & \downarrow \\ A(n) & = & A(1) & +(n-1) & \cdot d \end{array}$$

Explicit formula: $\quad A(n) = A(1) + (n-1)\,d$

For our sequence, $A(1) = 70$ and $d = 7$, so $A(n) = 70 + (n-1) \cdot 7$.

☛ A shorter way to see the same thing

Instead of adding all the equations, just substitute down the chain:

$$\begin{aligned} A(5) &= A(4) + 7 \\ &= \big[A(3) + 7\big] + 7 = A(3) + 2 \cdot 7 \\ &= \big[A(2) + 7\big] + 2 \cdot 7 = A(2) + 3 \cdot 7 \\ &= \big[A(1) + 7\big] + 3 \cdot 7 = A(1) + 4 \cdot 7 \end{aligned}$$

Count the steps, not the terms. Going from term 1 to term 5 takes **4** steps — which is exactly why the formula says $n - 1$ and not n.

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The two formulas side by side

	Recursive	**Explicit**
Formula	$A(1) = ____$ $A(n) = A(n-1) + d,\ n \geq 2$	$A(n) = A(1) + (n-1)d$
Says	"add d to the term before"	"start at $A(1)$, add d $(n-1)$ times"
Best for	the *next* term	a *faraway* term, like $A(100)$
Must include	**both** $A(1)$ and the rule	nothing else — it stands alone

✗ The three mistakes that cost the most points

1. **Writing (n) instead of $(n-1)$** in the explicit formula. Getting to term n takes $n-1$ steps. Test it: does your formula give the right *first* term when $n = 1$?
2. **Giving a recursive formula without $A(1)$.** "$A(n) = A(n-1) + 6$" describes infinitely many different sequences. **The starting value is part of the answer.**
3. **Getting the sign of d wrong on a decreasing sequence.** For $20, 17, 14, \ldots$ the common difference is -3, not 3. Always compute d as **later term minus earlier term**.

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Topic 1 · Is It Arithmetic? Find the Common Difference

Subtract consecutive terms and compare

How To Do It

Compute $A(2) - A(1)$, then $A(3) - A(2)$, then $A(4) - A(3)$.

- All the same ⇒ arithmetic, and that common value is d.
- Not all the same ⇒ not arithmetic.

Always subtract in the same direction: later term − earlier term.

☞ Worked Example

Is 70, 77, 84, 91, … **arithmetic? If so, find** d**.**

$$77 - 70 = 7, \qquad 84 - 77 = 7, \qquad 91 - 84 = 7$$

All three differences are equal, so

$$\boxed{\text{Yes — arithmetic, with } d = 7}$$

1. EASY

Is 5, 12, 19, 26, … arithmetic? If so, state d.

2. EASY *a decreasing sequence*

Is 20, 17, 14, 11, … arithmetic? If so, state d.

3. MEDIUM *fractions*

Is $\frac{1}{2}$, 1, $\frac{3}{2}$, 2, … arithmetic? If so, state d.

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4. MEDIUM *careful — this one is a trap*

Is $2,\ 4,\ 8,\ 16,\ \ldots$ arithmetic? Explain how you know.

5. HARD *solve for the unknown*

For what value of k is $3k-1,\ \ 2k+3,\ \ 5k-4$ an arithmetic sequence? Then state d.

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Topic 2 · The Recursive Formula

Where you start, and how you step

How To Do It

A recursive formula has **two parts**, and you need both:

$$A(1) = \text{first term} \qquad \text{and} \qquad A(n) = A(n-1) + d, \quad n \geq 2$$

For a **decreasing** sequence d is negative — write $A(n) = A(n-1) + (-8)$ or $A(n) = A(n-1) - 8$.

☞ Worked Example

Write the recursive formula for $3,\ 9,\ 15,\ 21,\ \ldots$

Step 1. First term $A(1) = 3$

Step 2. Second term $A(2) = A(1) + 6$

Step 3. Generalise $A(n) = A(n-1) + 6$

$$\boxed{A(1) = 3, \qquad A(n) = A(n-1) + 6, \quad n \geq 2}$$

6. EASY

Write the recursive formula for $4,\ 11,\ 18,\ 25,\ \ldots$

7. EASY *decreasing*

Write the recursive formula for $100,\ 92,\ 84,\ 76,\ \ldots$

8. MEDIUM *run the formula forwards*

A sequence is given by $A(1) = -4$ and $A(n) = A(n-1) + 9$ for $n \geq 2$. List the first five terms.

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9. MEDIUM *decimals*

Write the recursive formula for $2.5,\ 4,\ 5.5,\ 7,\ \ldots$

10. HARD

A sequence has $A(1) = 12$ and $A(n) = A(n-1) - 5$ for $n \geq 2$.

(a) List terms until you reach the first negative one.

(b) Which term is it? State both n and the value.

Topic 3 · The Explicit Formula

$A(n) = A(1) + (n-1)d$

How To Do It

$$A(n) = A(1) + (n-1)\,d$$

1. Find $A(1)$ and d.
2. Substitute them — **do not substitute** n.
3. Simplify by distributing, if the question asks for simplest form.
4. **Check with $n = 1$:** your formula must give back the first term.

☞ Worked Example

Write the explicit formula for $70,\ 77,\ 84,\ 91,\ \ldots$**, then simplify.**

Here $A(1) = 70$ and $d = 7$:

$$\begin{aligned} A(n) &= A(1) + (n-1)d \\ &= 70 + (n-1)\cdot 7 && \text{substitute} \\ &= 70 + 7n - 7 && \text{distribute} \\ &= 7n + 63 \end{aligned}$$

$$\boxed{A(n) = 70 + 7(n-1) = 7n + 63}$$

Check $n = 1$*:* $7(1) + 63 = 70$ ✓ *Check* $n = 3$*:* $21 + 63 = 84$ ✓

11. EASY

Write and simplify the explicit formula for $3,\ 9,\ 15,\ 21,\ \ldots$

12. EASY *decreasing*

Write and simplify the explicit formula for $20,\ 17,\ 14,\ 11,\ \ldots$

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13. MEDIUM *given $A(1)$ and d*

Write and simplify the explicit formula for the sequence with $A(1) = -7$ and $d = 4$.

14. MEDIUM *recursive → explicit*

A sequence is given recursively by $A(1) = 8$ and $A(n) = A(n-1) + 5$. Write its **explicit** formula in simplest form.

15. HARD *fractions, both formulas*

For $\frac{1}{4}, \frac{3}{4}, \frac{5}{4}, \frac{7}{4}, \ldots$ write **(a)** the recursive formula and **(b)** the explicit formula, simplified into a single fraction.

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Topic 4 · Finding a Term, Finding n

Forwards and backwards

How To Do It

Forwards — "find $A(20)$". Substitute $n = 20$ into the explicit formula.

Backwards — "which term equals 99?" Set the explicit formula *equal* to 99 and solve the equation for n.

> **n must come out a positive whole number.** If it does not, the value is not in the sequence.

☞ Worked Example

For 70, 77, 84, …, **find** $A(20)$.

$$\begin{aligned} A(20) &= A(1) + (20-1)d \\ &= 70 + 19 \cdot 7 \qquad \textbf{19} \text{ steps, not } 20 \\ &= 70 + 133 = 203 \end{aligned}$$

$$\boxed{A(20) = 203}$$

16. EASY

A sequence has $A(1) = 5$ and $d = 3$. Find $A(10)$.

17. EASY

For 4, 11, 18, 25, … find $A(12)$.

18. MEDIUM *negative d*

For 8, 5, 2, −1, … find $A(25)$.

19. MEDIUM *work backwards*

For 3, 9, 15, 21, ..., which term is equal to 99?

20. HARD *two terms given*

In an arithmetic sequence, $A(4) = 19$ and $A(9) = 44$.

(a) Find d. **(b)** Find $A(1)$. **(c)** Write the explicit formula.

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Topic 5 · Word Problems

Read the story, find $A(1)$ and d

How To Do It

Every arithmetic word problem gives you the same two things in disguise:

The starting amount	becomes $A(1)$
"each" / "every" / "per"	becomes d

Then decide which question is being asked: **find a term** (substitute n) or **find** n (solve an equation). **Answer in a sentence, with units.**

☞ Worked Example

A theater has 22 seats in the first row, and each row after has 3 more seats than the row before. How many seats are in row 15?

$A(1) = 22$ (first row) and $d = 3$ (each row after).

$$A(15) = 22 + (15 - 1) \cdot 3 = 22 + 14 \cdot 3 = 22 + 42 = 64$$

Row 15 has 64 seats.

21. EASY

Jaden deposits \$25 in week 1. Each week after, he deposits \$15 more than the week before. How much does he deposit in week 6?

22. EASY

A sapling is 40 cm tall when planted and grows 6 cm each year. Write an explicit formula for its height after n years, then find its height after 12 years.

23. MEDIUM

A gym charges a \$75 joining fee plus \$30 each month. Write a formula for the total paid after n months,

then find the total after 18 months.

24. MEDIUM *work backwards*

Logs are stacked with 30 in the bottom row, and each row above has 2 fewer logs. Which row has exactly 4 logs?

25. HARD

A company's profit was \$12,000 in year 1 and grew by the same amount each year. In year 7 the profit was \$30,000.

(a) Find d. **(b)** Write the explicit formula.

(b) In which year does the profit **first exceed** \$50,000?

Detailed Answer Key

Every step shown, every answer checked

Answers at a glance

1. Yes, $d = 7$

2. Yes, $d = -3$

3. Yes, $d = \frac{1}{2}$

4. No — differences $2, 4, 8$

5. $k = \frac{11}{4}$, $d = \frac{5}{4}$

6. $A(1) = 4$; $A(n) = A(n-1) + 7$

7. $A(1) = 100$; $A(n) = A(n-1) - 8$

8. -4, 5, 14, 23, 32

9. $A(1) = 2.5$; $A(n) = A(n-1) + 1.5$

10. $A(4) = -3$, the 4th term

11. $A(n) = 3 + 6(n-1) = 6n - 3$

12. $A(n) = 20 - 3(n-1) = 23 - 3n$

13. $A(n) = -7 + 4(n-1) = 4n - 11$

14. $A(n) = 8 + 5(n-1) = 5n + 3$

15. $A(n) = A(n-1) + \frac{1}{2}$; $\frac{2n-1}{4}$

16. $A(10) = 32$

17. $A(12) = 81$

18. $A(25) = -64$

19. the 17th term

20. $d = 5$, $A(1) = 4$, $A(n) = 5n - 1$

21. \$100

22. $H = 40 + 6n$; 112 cm

23. $T = 75 + 30n$; \$615

24. row 14

25. $d = 3000$; $3000n + 9000$; year 14

Topic 1 · Is It Arithmetic? Find d

1. $12 - 5 = 7$, $19 - 12 = 7$, $26 - 19 = 7$. All equal, so $\boxed{\text{yes, arithmetic with } d = 7}$.

2. $17 - 20 = -3$, $14 - 17 = -3$, $11 - 14 = -3$.

$$\boxed{\text{Yes, arithmetic with } d = -3}$$

The sign matters. The terms are going *down*, so d is negative.

3. $1 - \frac{1}{2} = \frac{1}{2}$, $\frac{3}{2} - 1 = \frac{1}{2}$, $2 - \frac{3}{2} = \frac{1}{2}$.

$$\boxed{\text{Yes, arithmetic with } d = \tfrac{1}{2}}$$

4. Compute the differences:

$$4 - 2 = 2, \qquad 8 - 4 = 4, \qquad 16 - 8 = 8$$

They are **not** all the same, so

$$\boxed{\text{No — it is not arithmetic.}}$$

What it *is*: each term is the one before *multiplied* by 2. That makes it a **geometric** sequence, not an arithmetic one.

✗ Common Mistake

Doubling "looks regular", which is exactly why this is a trap. **Arithmetic means you *add* the same amount; geometric means you *multiply* by the same amount.** Always check by subtracting.

5. (Hard) For the sequence to be arithmetic, the two gaps must be equal:

$$\begin{aligned}(2k+3)-(3k-1) &= (5k-4)-(2k+3) && \text{gap 1 = gap 2}\\ 2k+3-3k+1 &= 5k-4-2k-3 && \text{distribute both minus signs}\\ -k+4 &= 3k-7\\ 11 &= 4k \quad\Longrightarrow\quad \boxed{k=\tfrac{11}{4}}\end{aligned}$$

Now find d. Substituting $k=\frac{11}{4}$:

$$3k-1=\tfrac{29}{4},\qquad 2k+3=\tfrac{34}{4},\qquad 5k-4=\tfrac{39}{4}$$

$$d=\tfrac{34}{4}-\tfrac{29}{4}=\tfrac{5}{4},\qquad \tfrac{39}{4}-\tfrac{34}{4}=\tfrac{5}{4}\ \checkmark\qquad\Longrightarrow\qquad \boxed{d=\tfrac{5}{4}}$$

Topic 2 · The Recursive Formula

6. $d=11-4=7$, and the first term is 4.

$$\boxed{A(1)=4,\qquad A(n)=A(n-1)+7,\quad n\geq 2}$$

7. $d=92-100=-8$.

$$\boxed{A(1)=100,\qquad A(n)=A(n-1)-8,\quad n\geq 2}$$

8. Start at -4 and add 9 each time:

$$\begin{aligned}A(1)&=-4\\ A(2)&=-4+9=5\\ A(3)&=5+9=14\\ A(4)&=14+9=23\\ A(5)&=23+9=32\end{aligned}$$

$$\boxed{-4,\ 5,\ 14,\ 23,\ 32}$$

9. $d=4-2.5=1.5$.

$$\boxed{A(1)=2.5,\qquad A(n)=A(n-1)+1.5,\quad n\geq 2}$$

10. (Hard)

(a) Start at 12 and subtract 5 repeatedly:

$$12,\quad 7,\quad 2,\quad -3$$

(b) The first negative term is -3, and counting the positions $12=A(1)$, $7=A(2)$, $2=A(3)$, $-3=A(4)$:

$$\boxed{A(4)=-3 \text{ —the 4th term}}$$

Confirming with the explicit formula: $A(n) = 12 - 5(n-1) = 17 - 5n$, and

$$17 - 5n < 0 \implies n > \tfrac{17}{5} = 3.4 \implies \text{the first whole } n \text{ is } 4$$

$A(4) = 17 - 20 = -3$ ✓

Topic 3 · The Explicit Formula

11. $A(1) = 3$, $d = 6$:

$$A(n) = 3 + (n-1) \cdot 6 = 3 + 6n - 6 = \boxed{6n - 3}$$

Check $n = 1$: $6 - 3 = 3$ ✓ *Check* $n = 4$: $24 - 3 = 21$ ✓

12. $A(1) = 20$, $d = -3$:

$$A(n) = 20 + (n-1)(-3) = 20 - 3n + 3 = \boxed{23 - 3n}$$

Check $n = 1$: $23 - 3 = 20$ ✓ *Check* $n = 4$: $23 - 12 = 11$ ✓

13. $A(1) = -7$, $d = 4$:

$$A(n) = -7 + (n-1) \cdot 4 = -7 + 4n - 4 = \boxed{4n - 11}$$

Check $n = 1$: $4 - 11 = -7$ ✓

14. The recursive formula tells you both pieces directly: $A(1) = 8$ and $d = 5$.

$$A(n) = 8 + (n-1) \cdot 5 = 8 + 5n - 5 = \boxed{5n + 3}$$

Check: the sequence is $8, 13, 18, 23, \ldots$ and $5(3) + 3 = 18$ ✓

15. **(Hard)** First find d: $\frac{3}{4} - \frac{1}{4} = \frac{2}{4} = \frac{1}{2}$.

(a) Recursive:

$$\boxed{A(1) = \tfrac{1}{4}, \qquad A(n) = A(n-1) + \tfrac{1}{2}, \quad n \geq 2}$$

(b) Explicit:

$$\begin{aligned} A(n) &= \frac{1}{4} + (n-1) \cdot \frac{1}{2} \\ &= \frac{1}{4} + \frac{n}{2} - \frac{1}{2} && \text{distribute} \\ &= \frac{n}{2} - \frac{1}{4} && \tfrac{1}{4} - \tfrac{1}{2} = -\tfrac{1}{4} \\ &= \frac{2n}{4} - \frac{1}{4} = \boxed{\frac{2n-1}{4}} \end{aligned}$$

Check $n = 1$: $\frac{1}{4}$ ✓ *Check* $n = 4$: $\frac{7}{4}$ ✓

☛ Note

The simplified form $\dfrac{2n-1}{4}$ says something nice: the numerators $1, 3, 5, 7, \ldots$ are the **odd numbers**, and $2n - 1$ is the standard formula for the nth odd number.

Topic 4 · Finding a Term, Finding n

16. $A(10) = 5 + (10-1)\cdot 3 = 5 + 27 = \boxed{32}$

17. $d = 7$, so $A(12) = 4 + (12-1)\cdot 7 = 4 + 77 = \boxed{81}$

18. $d = 5 - 8 = -3$, so

$$A(25) = 8 + (25-1)(-3) = 8 - 72 = \boxed{-64}$$

24 steps, not 25. That single subtraction is where most of the errors happen.

19. From Problem 11, $A(n) = 6n - 3$. Set it equal to 99 and solve:

$$\begin{aligned} 6n - 3 &= 99 \\ 6n &= 102 \\ n &= 17 \end{aligned}$$

$$\boxed{99 \text{ is the 17th term}}$$

Check: $A(17) = 6(17) - 3 = 102 - 3 = 99$ ✓

20. (Hard)

(a) Find d. Going from term 4 to term 9 is **5 steps**, and the value rose by $44 - 19 = 25$:

$$5d = 25 \quad \Longrightarrow \quad \boxed{d = 5}$$

(b) Find $A(1)$. Use $A(4) = A(1) + 3d$:

$$\begin{aligned} 19 &= A(1) + 3(5) \\ 19 &= A(1) + 15 \quad \Longrightarrow \quad \boxed{A(1) = 4} \end{aligned}$$

(c) Explicit formula.

$$A(n) = 4 + (n-1)\cdot 5 = 4 + 5n - 5 = \boxed{5n - 1}$$

Check both given values: $A(4) = 20 - 1 = 19$ ✓ $A(9) = 45 - 1 = 44$ ✓

> ☛ **Note**
>
> **The count of steps is the whole trick.** From term 4 to term 9 is $9 - 4 = 5$ steps — not 4, not 6. In general $A(q) - A(p) = (q - p)\,d$, which is a fast way to find d from any two terms.

Topic 5 · Word Problems

21. $A(1) = 25$ (week 1) and $d = 15$ (each week after).

$$A(6) = 25 + (6-1)\cdot 15 = 25 + 75 = 100$$

Jaden deposits $100 in week 6.

22. The tree is 40 cm at planting, and gains 6 cm per year, so after n years

$$\boxed{H = 40 + 6n}$$

After 12 years: $H = 40 + 6(12) = 40 + 72 = 112$. **The tree is 112 cm tall after 12 years.**

☛ Note

Notice this one uses $6n$, **not** $6(n-1)$ — because n counts *years of growth*, and at $n=0$ (planting time) the height is 40. **Always ask what $n=1$, or $n=0$, is supposed to mean in the story** before choosing the formula.

23. The \$75 is paid once at the start; the \$30 repeats each month:

$$\boxed{T = 75 + 30n}$$

After 18 months: $T = 75 + 30(18) = 75 + 540 = 615$. **The total paid is \$615.**

24. $A(1) = 30$ (bottom row) and $d = -2$ (each row above). Set the formula equal to 4:

$$\begin{aligned} 30 + (n-1)(-2) &= 4 \\ 30 - 2n + 2 &= 4 \\ 32 - 2n &= 4 \\ 28 &= 2n \qquad \Longrightarrow \qquad n = 14 \end{aligned}$$

Row 14 has exactly 4 logs. *Check:* $30 + 13(-2) = 30 - 26 = 4$ ✓

25. (Hard)

(a) Find d. From year 1 to year 7 is **6 steps**, and profit rose by $30{,}000 - 12{,}000 = 18{,}000$:

$$6d = 18{,}000 \qquad \Longrightarrow \qquad \boxed{d = 3{,}000}$$

(b) Explicit formula.

$$A(n) = 12{,}000 + (n-1)(3{,}000) = 12{,}000 + 3{,}000n - 3{,}000 = \boxed{3{,}000n + 9{,}000}$$

Check: $A(7) = 21{,}000 + 9{,}000 = 30{,}000$ ✓

(c) First year exceeding \$50,000. Solve the inequality:

$$\begin{aligned} 3{,}000n + 9{,}000 &> 50{,}000 \\ 3{,}000n &> 41{,}000 \\ n &> \frac{41}{3} = 13.67\ldots \end{aligned}$$

Since n counts whole years, the first one that works is $n = 14$:

$$\boxed{\text{Year } 14}$$

Check both sides of the boundary:

Year 13	$A(13) = \$48{,}000$	not more than \$50,000
Year 14	$A(14) = \$51{,}000$	first year above \$50,000 ✓

✗ Common Mistake

$n > 13.67$ does **not** round to 14 — it is *cut up* to 14. Rounding 13.67 would also give 14 here by luck, but if the answer had been $n > 13.2$ the correct year would still be 14, while rounding would wrongly give 13. **When n must be a whole number, always take the next whole number**

up, then check it.

The six things to remember

1. **Arithmetic means you ADD the same number each time.** Test by subtracting consecutive terms — later minus earlier.
2. **A recursive formula needs both parts:** the starting value $A(1)$ *and* the rule $A(n) = A(n-1) + d$ for $n \geq 2$.
3. **The explicit formula is** $A(n) = A(1) + (n-1)d$. It exists because getting to term n takes $n-1$ steps of size d.
4. **Check every formula at** $n = 1$. It must give back the first term.
5. **Going from term** p **to term** q **is** $q - p$ **steps**, so $A(q) - A(p) = (q-p)d$. That finds d from any two terms.
6. **"Which term is...?" means solve an equation for** n, and n must be a positive whole number.

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